



INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO



Selected Topics in Cryptography

Session 1: Elliptic curves

September 8, 2026

Please do the following exercises by your own. Choose only one of the following programming languages to develop your code: C/C++, Python or Java.

1. Preliminary functions

At the end of the session you must upload a pdf file, containing screenshots of your source code, and screenshots showing your programs running.

1. Design and implement a function that receives as a parameter a prime number $p > 3$. Your function must print the quadratic residues module p (QR_p) and for each quadratic residue print its square roots. Assume that p is a prime number, you do not need to verify that it is a prime number.
2. Design and implement a function that receives as parameters a prime number $p > 3$, the parameters of a non-singular elliptic curve $a, b \in \mathbb{Z}_p$. Then use the function you designed in the previous exercise to find and count the rational-points of the elliptic curve. Represent any point with three coordinates (x, y, z) , $z = 1$ for every point different from infinity point, and $(0, 1, 0)$ for the point to infinity. Store in a text file the values p, a, b and the set of rational points of the elliptic curve you found.
3. Design and implement a program in such a way that we can test each function separately.

2. Elliptic curve arithmetic

To develop the following exercises you can use a library of the programming language to do arithmetic with big integers. Consider that every parameter such as a, b, p and the coordinates of the point in an elliptic curve could be a big integer. Use an existing function in the programming language to generate a prime number.

1. Design and implement a function to find a non-singular elliptic curve in a finite field $GF(p)$. Your function must receive the length in bits n of a prime number p . Then it must randomly generate a prime number of n bits and randomly find the values a, b of a non-singular elliptic curve $y^2 = x^3 + ax + b \pmod{p}$.
2. Design and implement a function to do point addition, i.e. $P + Q$, where $P \neq \pm Q$. Verify that P and Q belong to the given elliptic curve $y^2 = x^3 + ax + b \pmod{p}$. Please use three coordinates to represent each point. Your function must receive as parameters a, b, p, P, Q and must return the result of the addition point.
3. Design and implement a function to do point doubling, i.e. $2P$. Verify that P belongs to the given elliptic curve $y^2 = x^3 + ax + b \pmod{p}$. Please use three coordinates to represent P . Your function must receive as parameters a, b, p, P and must return the result of point doubling.
4. Design and implement a program to test each of the previous function separately.

3. Questions

Answer the following questions:

1. For three different non-singular elliptic curves in a finite field $GF(p)$, where $3 \leq p < 10,000$. For each prime number, use the text file generated in the exercise 2 of Section 1 and do the graph of the rational points. To do this graph you can use any software available.
2. Generate a prime number for each size: 16 bits, 32 bits, 64 bits, 512 bits, 1024 bits and 2048 bits. For each prime number, find a non-singular elliptic curve.
3. For each of the following set of values, compute $P + Q, 2P, 2Q$
 - a) $E : y^2 = x^3 + x + 1 \pmod{65537}, P = (49606, 64426, 1), Q = (2565, 62370, 1)$
 - b) $E : y^2 = x^3 + 30x + 97 \pmod{4294967311}, P = (433318550, 1866632789, 1), Q = (408186704, 4022951807, 1)$
 - c) $E : y^2 = x^3 + 125x + 2 \pmod{2^{31}-1}, P = (1506532484, 1041296099, 1), Q = (1624813594, 253477454, 1)$
 - d) $E : y^2 = x^3 + x + 1300 \pmod{2^{61}-1}, P = (1317571598731990128, 494998261481053431, 1), Q = (590181223958911612, 1863749232038030155, 1)$

4. Products

Every student must write a brief report, containing:

1. Your personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code you wrote to solve exercises 1 and 2 of Section 1, and a brief explanation of how it work.
3. Source code you wrote for each function to solve exercises 1-3 of Section 2, and a brief explanation of how it work.
4. Answers to questions of Section 3.
5. If you use any AI for some parts of your code, please give the name of AI you used. Also you must indicate what source code was given by the AI. Justify why you need to use it.

You must upload the pdf file of your report to Microsoft Teams.